

Assignment 2

Name: Shivam Gupta

Description

- **Generate Book Data:**
 - Create a list of 100 random books (bookId, genre, price, author, publicationYear) and store them in the local file as JSON(AWS S3).
 - Include additional fields: ISBN, publisher, language, and cover image URL
- **Advanced Search:**
 - Implement **fuzzy search** feature that allows filtering books by multiple criteria (bookId, author).
 - Include date range picker for publication year.
 - Also, include a range picker for the price of the book.
 - Implement tag-based search for genres (allow multiple genre selections)
- **Sorting and Pagination:**
 - Implement sorting functionality that allows sorting by multiple fields (price, publicationYear) in ascending/descending order.
 - Implement custom pagination (5, 10, 25 books per page).
- **Book Details:**
 - When examining a book, show detailed information including author and publication year. (a modal view can be created)
 - Display similar books based on the genre and price range (+/- 10%).
- **Data Persistence:**
 - Use Local Storage to persist the state of the book list, so the data remains even after refreshing the page.
- **Error Handling:**
 - Implement robust error handling for the file reading/writing operations and API requests.
- **Promises and Async/Await:**
 - Complete the assignment using both Promises and Async/Await separately.
 - Create two versions of the solution: one using Promises and one using Async/Await.
- **User Interface Enhancements:**
 - Create a responsive UI.
 - Implement dark/light mode toggle.
- **Performance Optimization:**
 - Implement **lazy loading** for book cover images.
 - Use virtual scrolling for large lists to improve performance.
- **Code Quality and Best Practices:**
 - Follow ES6 syntax and industry-standard coding practices.

- Use Git for version control and maintain best practices while committing the code to GitHub.

Purpose of this Assignment:

- Gain a deeper understanding of Asynchronous Programming in JavaScript.
- Work with more advanced ES6+ features.
- Enhance skills in implementing complex search and sort functionalities.
- Practice pagination and state persistence with Local Storage.
- Use Git for version control and follow best practices for code commits.

Note: See the HTML code below that will incorporate all the abovementioned features. You need to build the backend section that supports the functionality discussed above.

HTML Code

Please note that it's a basic layout and extra functionalities must be added to the UI itself.

```
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <meta http-equiv="X-UA-Compatible" content="ie=edge">
    <link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.1.3/css/bootstrap.min.
css" integrity="sha384-
MCw98/SFnGE8fJT3GXwEOngsV7Zt27NXFoaoApmYm81iuXoPkFOJwJ8ERdknLPMO"
crossorigin="anonymous">

    <title>Book Catalog</title>
  </head>

  <body>
    <div id="div-search" class="container" style="margin-bottom:20px;
padding-top:20px;">
      <h3>Search Books</h3>
      <form action="#">
        <input type="text" id="input-search-id" class="form-
control" style="width:25%;display:inline;" placeholder="Book Id">
        <input type="text" id="input-search-genre" class="form-
```

```

control" style="width:25%;display:inline;" placeholder="Genre">
    <input type="text" id="input-search-price-min" class="form-
control" style="width:12%;display:inline;" placeholder="Price Min">
    <input type="text" id="input-search-price-max" class="form-
control" style="width:12%;display:inline;" placeholder="Price Max">
    <input type="text" id="input-search-author" class="form-
control" style="width:25%;display:inline;" placeholder="Author">
    <input type="text" id="input-search-year" class="form-
control" style="width:25%;display:inline;" placeholder="Publication Year">
    <button id="btn-search" class="btn"
type="submit">Search</button>
    </form>
</div>

    <div id="div-examined-book" class="container" style="margin-
bottom:20px">
        <h3>Examined Book</h3>
        <ul id="list-examined-book" class="list-group">
            <li class="list-group-item">Book Id: <span id="book-
id"></span></li>
            <li class="list-group-item">Price: <span id="book-
price"></span></li>
            <li class="list-group-item">Genre: <span id="book-
genre"></span></li>
            <li class="list-group-item">Author: <span id="book-
author"></span></li>
            <li class="list-group-item">Publication Year: <span
id="book-year"></span></li>
        </ul>
    </div>

    <div class="container" style="margin-bottom:20px">
        <h3>List of Similar Books</h3>
        <table id="table-similar-books" class="table">
            <thead>
                <th>BookId</th>
                <th>Genre</th>
                <th>Price</th>
                <th>Author</th>
                <th>Publication Year</th>
                <th>Examine</th>
            </thead>
            <tbody></tbody>

```

```

        </table>
    </div>

    <div class="container" style="margin-bottom:20px">
        <h3>List of All Books</h3>
        <table id="table-all-books" class="table">
            <thead>
                <th>BookId</th>
                <th>Genre</th>
                <th>Price</th>
                <th>Author</th>
                <th>Publication Year</th>
                <th>Examine</th>
            </thead>
            <tbody></tbody>
        </table>

        <nav aria-label="Page navigation">
            <ul class="pagination">
                <li class="page-item"><a class="page-link"
href="#">Previous</a></li>
                <li class="page-item"><a class="page-link"
href="#">1</a></li>
                <li class="page-item"><a class="page-link"
href="#">2</a></li>
                <li class="page-item"><a class="page-link"
href="#">3</a></li>
                <li class="page-item"><a class="page-link"
href="#">Next</a></li>
            </ul>
        </nav>
    </div>

</body>
</html>

```